

quiz

```
x <- c(1:7,NA)
x1 <- ifelse(is.na(x),1,0)
x1
```

```
## [1] 0 0 0 0 0 0 0 1
```

```
y <- c(1.29, 2.35, 2.95, 3.77, 5.30, 5.09, 4, 4.5)
cbind(x,y)
```

```
##      x    y
## [1,] 1 1.29
## [2,] 2 2.35
## [3,] 3 2.95
## [4,] 4 3.77
## [5,] 5 5.30
## [6,] 6 5.09
## [7,] 7 4.00
## [8,] NA 4.50
```

```
# Impute
x[is.na(x)] <- sum(1 +2 +3 +4 +5 +6 +7)/8
x
```

```
## [1] 1.0 2.0 3.0 4.0 5.0 6.0 7.0 3.5
```

```
# fit regression
```

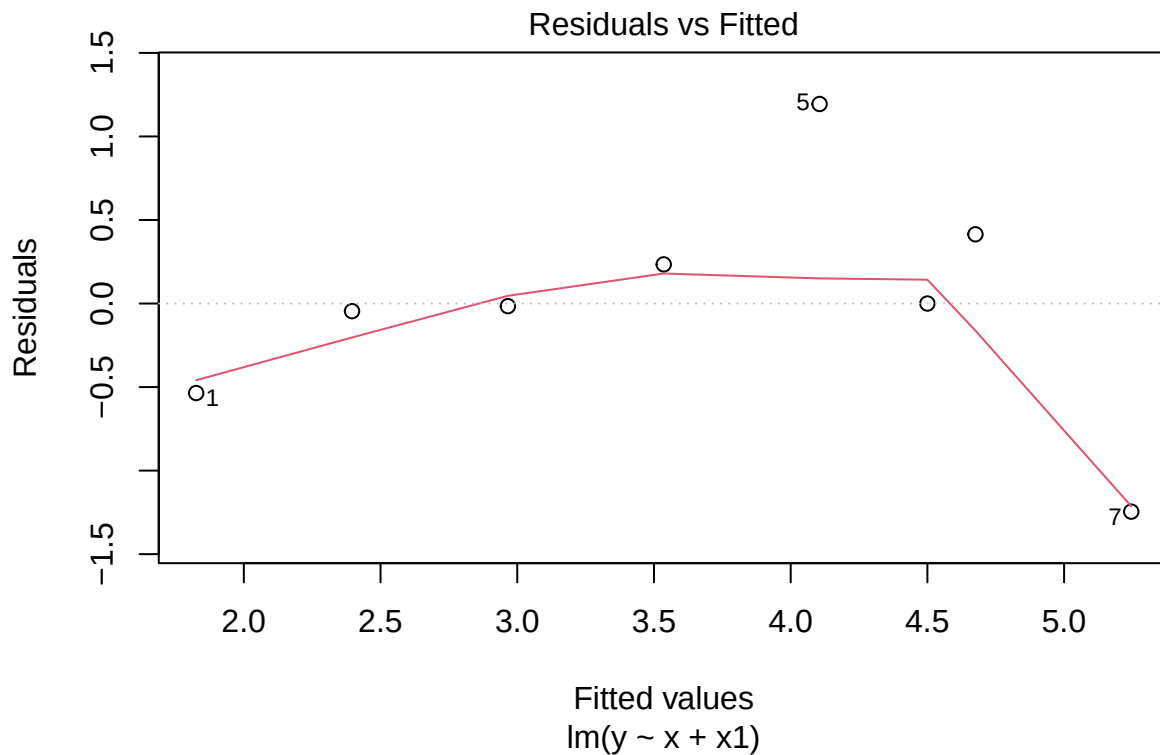
```
m1 <- lm(y ~ x + x1)
summary(m1)
```

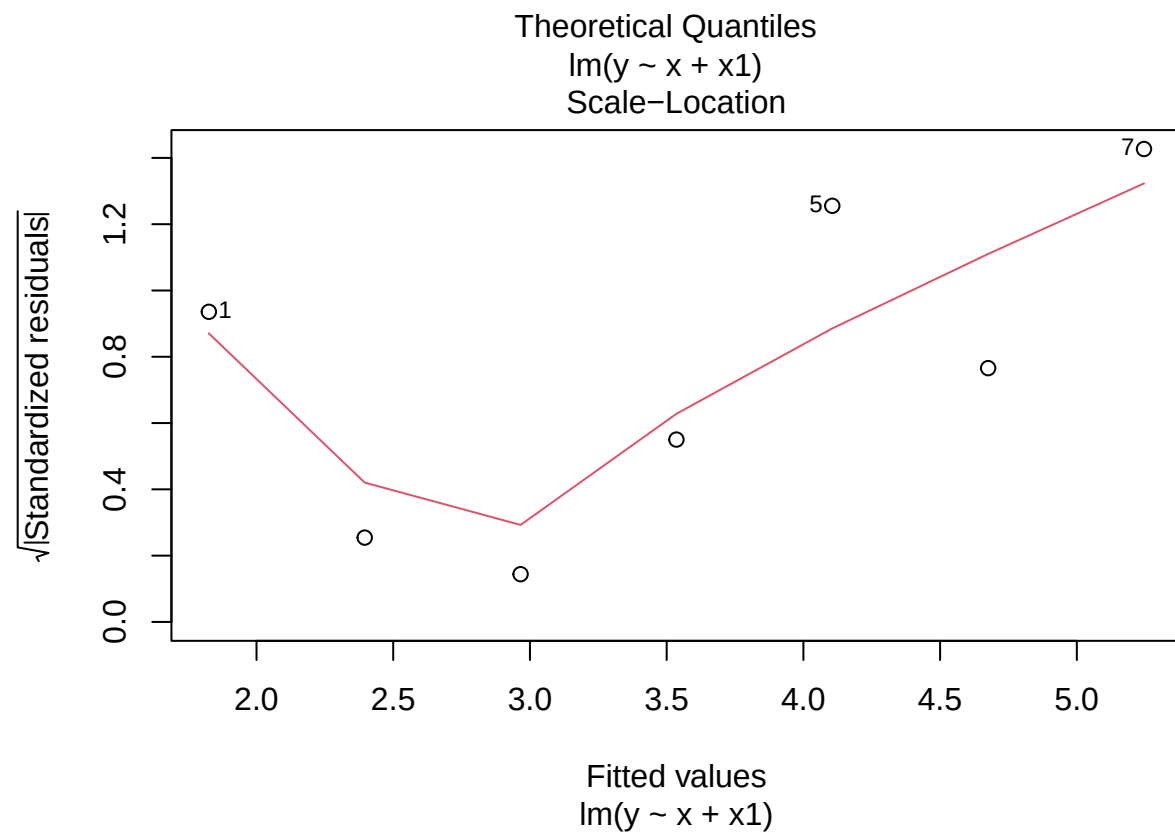
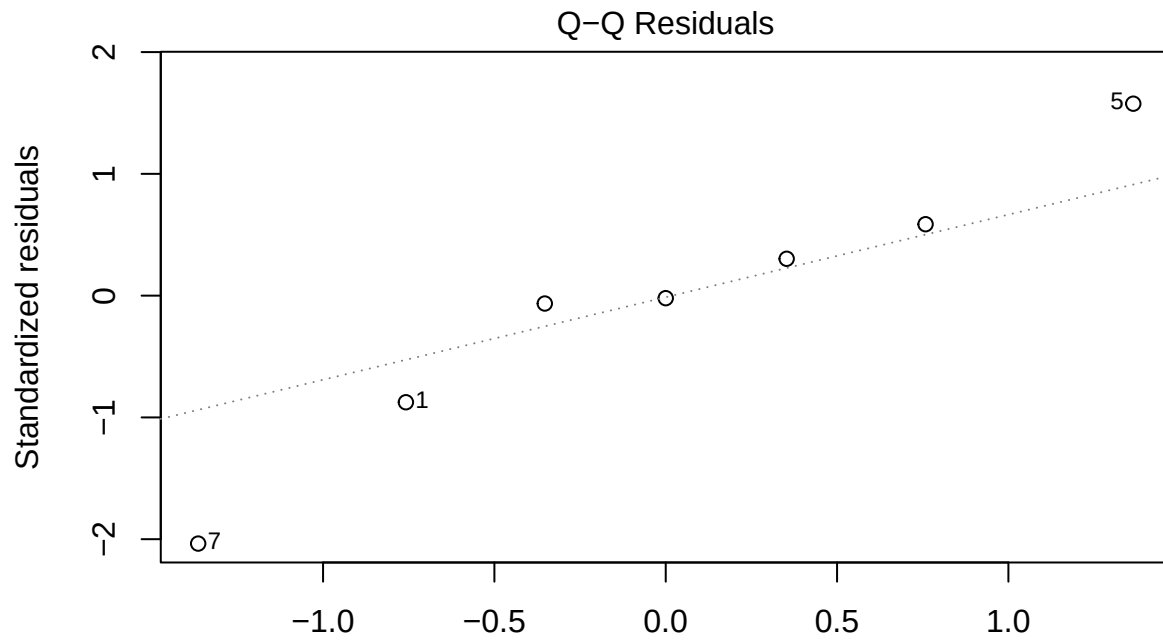
```
##
## Call:
## lm(formula = y ~ x + x1)
##
## Residuals:
##      1      2      3      4      5      6      7
## -5.357e-01 -4.571e-02 -1.571e-02  2.343e-01  1.194e+00  4.143e-01 -1.246e+00
##      8
##  2.732e-17
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

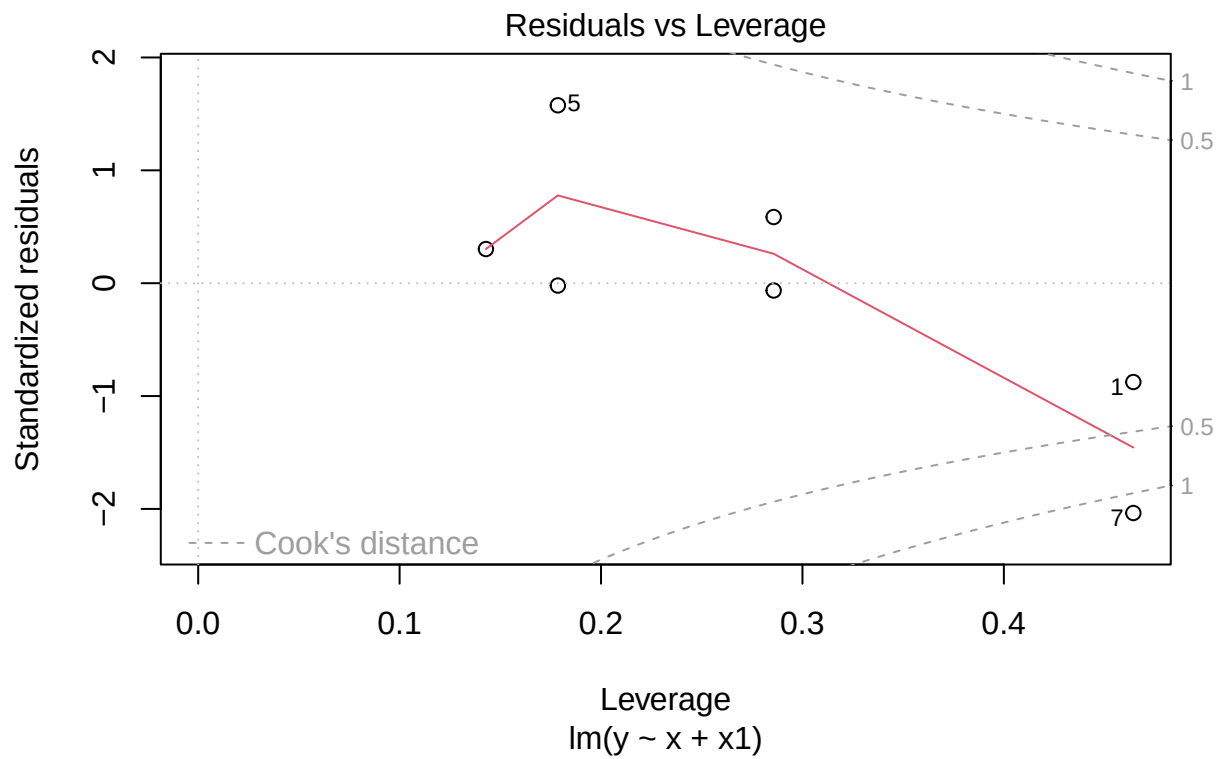
```
## (Intercept)  1.2557    0.7065    1.777    0.1357
## x           0.5700    0.1580    3.608    0.0154 *
## x1          1.2493    0.8971    1.393    0.2225
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8359 on 5 degrees of freedom
## Multiple R-squared:  0.7393, Adjusted R-squared:  0.6351
## F-statistic: 7.091 on 2 and 5 DF,  p-value: 0.03469
```

```
plot(m1)
```

```
## Warning: not plotting observations with leverage one:
##      8
```







```
m1$fitted.values
```

```
##      1      2      3      4      5      6      7      8
## 1.825714 2.395714 2.965714 3.535714 4.105714 4.675714 5.245714 4.500000
```

```
m1$residuals
```

```
##      1      2      3      4      5
## -5.357143e-01 -4.571429e-02 -1.571429e-02 2.342857e-01 1.194286e+00
##      6      7      8
## 4.142857e-01 -1.245714e+00 2.732244e-17
```

```
plot(x,y)
abline(x,y)
```

